F23PP468-VS
VARIABLE SPEED DRIVE
SYSTEM INTERCONNECTION DIAGRAM DRG. No. 2000-EL005-B02-10002

11,000V
RD WH BU
FCF 160A
QAC 160A 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
SWITCHBOARD EARTH BAR
NOTE 3
SUB FLOOR DROP BOX
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE
EARTH BAR
TRANSFORMER EARTH BAR
L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10011 SHEET 14

OUTGOING MOTOR SUPPLY SHEET 6
T1 T2 T3
CABLE SCREEN EARTHED
M 3~
F23-PP-468
FINAL TAILS PUMP 3
3,300V
800 kW

MOTOR SPACE HEATER CIRCUIT SHEET 6
X2 54 53
F23PP468HT-L
F23PP468HT-N
F23-PP-468
MOTOR HEATER TERMINAL BOX
2000-ME063B-F14-10013
1 (H1) 3 (H2) 0.3kW
ANTI-CONDENSATION HEATER

A80 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6
X81
PT100 RTD INPUT CH1
PT100 RTD INPUT CH2
PT100 RTD INPUT CH3
PT100 RTD INPUT CH4
PT100 RTD INPUT CH5
PT100 RTD INPUT CH6
PT100 RTD INPUT CH7
PT100 RTD INPUT CH8
COM
F23PP468A-RD
F23PP468A-WH
F23PP468A-BK
F23PP468B-RD
F23PP468B-WH
F23PP468B-BK
F23PP468C-RD
F23PP468C-WH
F23PP468C-BK
F23PP468D-RD
F23PP468D-WH
F23PP468D-BK
F23PP468E-RD
F23PP468E-WH
F23PP468E-BK
F23-PP-468
MOTOR RTD TERMINAL BOX
2000-ME063B-F14-10013
RTD
MOTOR WINDING RTD (RD PHASE) (TD1)
MOTOR WINDING RTD (WH PHASE) (TD3)
MOTOR WINDING RTD (BU PHASE) (TD5)
MOTOR BEARING RTD (DRIVE END) (BD10)
MOTOR BEARING RTD (NON DRIVE END) (BD20)
FIELD CABLE TERMINALS

F23-SW-016 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10011 SHEET 7 3A & 3B
X3 1139 3
X2 67
X3 2713 17
X2 68
K41 K41
MV CONTACTOR ENABLE CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10011 SHEET 10 4A
X2 4 1179
X2 63
K41 K41 K41
X2 25 1199
X2 64
MV INPUT ENABLE CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER VSD TRIP 2000-EL003-F16-10011 SHEET 10 3A
X2 3 1219
X2 65
X2 24 1239
X2 66
K41 K41 K41
MV INPUT TRIP CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10011 SHEET 8 5A & 5D
X2 56 1259
X2 70
X2 44 1279
X2 69
A30 (PART OF)
X2 15 TRIP X2 16
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F23-DB-101
2000-F23-TLD-EL-10054
RD WH BU BK
R4 W4 B4
25A
415VAC
EARTH BAR
F23DB101-R4
F23DB101-W4
F23DB101-B4
F23DB101-BK4
X4 1 2 3 N PE1
VSD AUXILIARY POWER SUPPLY SHEET 6

F23-DB-201
2000-F23-TLD-EL-10101
RD BK
Q175
10A
240VAC
EARTH BAR
F23DB201-Q175 (RD)
F23DB201-Q175 (BK)
X4 5 6 PE2
VSD UPS POWER SUPPLY SHEET 6

TO FINAL TAILS PUMPS
415V SWITCHROOM
F23-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F23-ETH-001
2000-F23-TER-IC-10003
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47
ETHERNET ADAPTOR RJ45
SHEET 6

A2 (PART OF)
DI-2B 9
X2 13
24VDC CONTROL SHEET 6
14

SAFE TORQUE OFF CIRCUIT SHEET 6
X86 1 2 3 4
F23PP468VS-1409
F23PP468VS-1419
F23PP468VS-1429
F23PP468VS-1439
F23PP468VS-1459
F23PP468VS-1469
F23PP468-LCS
LOCAL CONTROL STATION
X1 1 2 3 4 5 6 E
LOS
START

## NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F23PP468-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
3. IF THE CABLE SCREENS PASS THROUGH CORE BALANCE CT BEFORE BREAKING OUT THEN THE SCREEN EARTHS MUST BE BOUGHT BACK THROUGH THE CORE BALANCE CT AND TERMINATED TO EARTH.

## LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A05

# 2000-F23-SCM-EL-10109_Rev1

F23-SW-016 - F23-PP-468 final tails pump 3 VSD feeder - schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F23 - FINAL TAILINGS HANDLING & STORAGE\2000-F23-SCM-EL-10109.DWG 02OCT25 9:03AM HOTSEAT04